

Alex Pawelczyk

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PROFESSIONAL SUMMARY

Full-stack software engineer specialized in building and modernizing cloud-native applications using Java, Spring Boot, Angular, and Azure-native technologies. Experienced developing scalable backend services, secure API integrations, and intuitive user interfaces within large, business-critical environments. Proven track record of delivering production-ready solutions that align technical design with operational and business needs, while improving engineering quality through automation, testing, and collaborative development practices. Strong expertise in designing and deploying AI-powered systems using LangGraph, Glean, and LLM-based workflows.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python

Frameworks & Development: Spring Boot, Angular, React Native, Node.js

Cloud & Infrastructure: Azure, Kubernetes/OpenShift, Docker

Databases: PostgreSQL, Oracle, SQL Server, Redis, ChromaDB

DevOps & CI/CD: GitHub Actions, Azure DevOps, SonarQube, CodeQL, CI/CD

Observability & Quality: Datadog, OpenTelemetry, JUnit, Playwright, TDD

API & Security: REST APIs, OAuth 2.0, Azure APIM, Azure Key Vault, Hashi Vault

AI & Developer Tooling: LangGraph, LangChain, Azure AI, GitHub Copilot, Glean

Tools & Collaboration: Git, JIRA, Agile/Scrum, Code Review

PROFESSIONAL EXPERIENCE

Lead Software Engineer

General Motors, Warren, MI | Mar 2021 – May 2026

- Led modernization and migration initiatives for enterprise applications on Azure AKS and Red Hat OpenShift using containerized microservices architectures, including multiple successful production migrations of critical business systems.
- Designed and implemented AI-enabled engineering workflows using Glean and retrieval-augmented generation (RAG) techniques to automate operational and development processes.
- Engineered secure CI/CD pipelines with GitHub Actions and JFrog Artifactory, integrating automated quality and security tooling including SonarQube, CodeQL, and Vault.
- Developed and deployed Spring Boot microservices supporting enterprise workforce and badge management integrations during a phased UKG migration, leveraging Azure PostgreSQL, Key Vault, OAuth 2.0, and JWT-based authentication.
- Established testing standards and automation frameworks using JUnit, Mockito, and Playwright, contributing to a significant increase in code coverage and engineering quality across multiple applications.
- Led development and ongoing enhancement of a global enterprise vehicle management platform built with Angular and Spring Boot, supporting a large-scale application ecosystem with 100+ UI screens and distributed services used across the organization.

- Collaborated closely with product, design, and engineering teams in Agile environments to deliver scalable, production-ready software aligned with business objectives.
- Mentored junior engineers and interns through code reviews, pair programming, and technical guidance, helping improve development quality and team effectiveness.

EDUCATION

Master of Mathematics in Computer Science

University of Waterloo — Waterloo, ON | Sep 2019 – Dec 2020

GPA: 3.70 | U Sports Academic All-Canadian | Varsity Athletic Scholarship

Bachelor of Computer Science

Dalhousie University — Halifax, NS | Sep 2016 – Aug 2019

GPA: 3.67 | U Sports Academic All-Canadian (2x) | Sexton Scholar (3x) | Dean's List | Varsity Athletic Scholarship (3x)

PUBLICATIONS

- D. T. Radke, D. L. Radke, T. Brecht, and A. Pawelczyk. Passing and pressure metrics in ice hockey. Artificial Intelligence for Sports Analytics Workshop at IJCAI (AISA-IJCAI 2021), 2021.
cs.uwaterloo.ca/~dtradke/aisa-hockey-paper.html

PROJECTS

Agentic Chat Bot | github.com/apawel63/agentic-chatbot

- Built a multi-purpose AI chatbot application that uses LangGraph to orchestrate complex AI workflows as state graphs, allowing for sophisticated multi-step processes with conditional logic, tool integration, and message management.

Local AI Agent with RAG | github.com/apawel63/local-ai-agent-with-rag

- Built a local AI-powered RAG chatbot that answers pizza restaurant questions using semantic search over customer reviews with vector databases and LLMs.